

Scientific Analysis of the SON Coronal Mass Ejection Sonification Experiment

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Abstract

This report explains the SON coronal mass ejection (CME) experiment as a reduced-order scientific sonification and visualization system. The supplied source code defines an event database of solar-wind and geomagnetic disturbances, a physics module that computes diagnostic quantities from event-level inputs, a sonification module that maps those diagnostics to multiple audio layers, and a visualization module that scales boundary geometry and particle motion from the same state variables. The analysis shows that the experiment is not a first-principles magnetohydrodynamic simulation. Instead, it is an empirical and phenomenological encoding pipeline that combines measured or reconstructed event parameters with established space-physics relations, heuristic scaling laws, and browser-based audio rendering. The report documents the formulas implemented in the code, the role of the major functions, the sonification architecture, the visualization logic, and the limitations that arise from simplified assumptions, unit normalizations, and heuristic scaling choices.

Keywords: coronal mass ejection, space weather, sonification, magnetosphere, Web Audio, empirical model

Method and System Definition

The supplied SON source defines a browser-based experiment in which a discrete CME event record is transformed into both sound and image through a sequence of computational functions. At the input stage, each event is represented by a parameter object p containing solar-

wind speed v (km/s), proton number density n (cm^{-3}), interplanetary magnetic-field magnitude B (nT), southward field component B_z (nT), geomagnetic planetary index K_p , storm-time disturbance index Dst (nT), flare-associated X-ray proxy x_{ray} (W/m^2), and event duration dur (h). The event records also carry a confidence label (direct, partial, or historical), which explicitly separates direct spacecraft measurements from partially reconstructed or historical estimates.

The experiment can therefore be described as a two-stage transformation. First, the function `computePhysics(p)` converts the event-state vector into a set of derived physical diagnostics. Second, the function `buildGraph(ctx, p, ph)` maps the original parameters p and the derived physics object ph into a multi-layer audio graph implemented through the Web Audio API. A parallel visualization branch uses the same derived state variables to scale the magnetopause, bow shock, magnetotail, and particle trajectories on a canvas.

Table 1. Input variables encoded in the event-state vector.

Symbol	Code key	Units	Meaning
v	<code>p.v</code>	km/s	Bulk solar-wind speed
n	<code>p.n</code>	cm^{-3}	Proton number density
B	<code>p.B</code>	nT	Interplanetary magnetic-field magnitude
B_z	<code>p.Bz</code>	nT	North-south magnetic-field component
K_p	<code>p.Kp</code>	index	Planetary geomagnetic activity index
Dst	<code>p.Dst</code>	nT	Storm-time disturbance index
x_{ray}	<code>p.xray</code>	W/m^2	Flare-associated X-ray proxy
dur	<code>p.dur</code>	h	Nominal event duration
confidence	confidence	category	Measurement/reconstruction quality

			label
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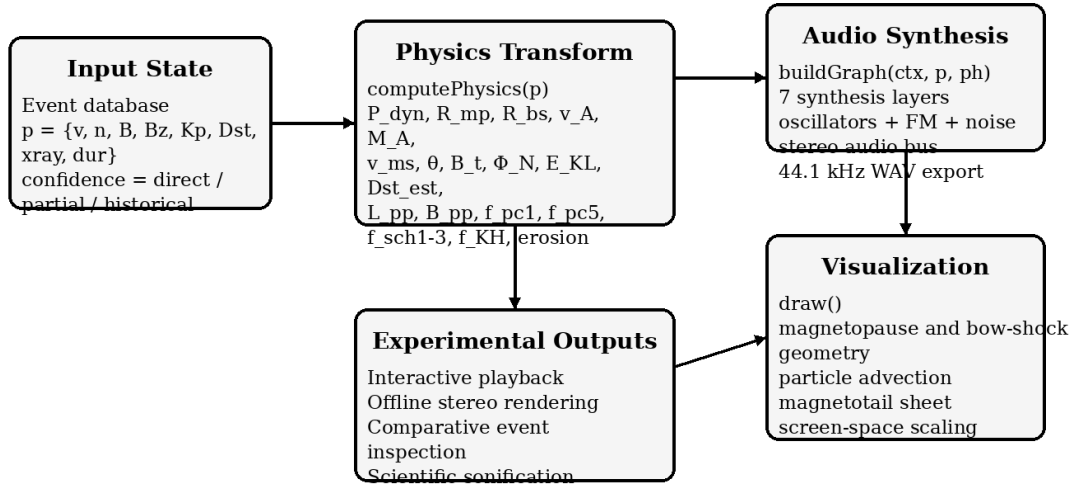


Figure 1. Computational pipeline of the SON experiment.

Implemented Physics Functions

The core scientific function is `computePhysics(p)`. This function converts the event parameters into derived quantities that summarize compression of the magnetosphere, energy-coupling potential, storm-time ring-current response, plasmaspheric displacement, and characteristic wave frequencies. The equations below are reproduced in the form implemented by the source code.

Solar-wind dynamic pressure.

$$P_{\text{dyn}} = 0.5 n m_p v^2$$

The code converts density from cm^{-3} to m^{-3} and speed from km/s to m/s , computes kinetic pressure in pascals, and rescales the result to nPa.

Magnetopause stand-off distance.

$$R_{\text{mp}} = 11.4 P_{\text{dyn}}^{(-1/6.6)}$$

An empirical pressure law expresses dayside compression as pressure increases.

Bow-shock stand-off distance.

$$R_{bs} = 1.36 R_{mp}$$

The shock is positioned at a fixed multiple of the magnetopause distance.

Alfvén speed.

$$v_A = B / \sqrt{\mu_0 n m_p}$$

This gives the characteristic MHD wave speed implied by field strength and mass density.

Alfvénic Mach number.

$$M_A = v / v_A$$

This dimensionless quantity measures how strongly the flow exceeds the Alfvén speed.

Magnetosonic speed.

$$v_{ms} = \sqrt{v_A^2 + (0.4 v)^2}$$

The code uses a fixed sound-speed proxy and combines it quadratically with the Alfvén speed.

IMF clock angle.

$$\theta = \arccos(-B_z / B)$$

The clock angle controls reconnection efficiency and later coupling relations.

Transverse IMF.

$$B_t = \sqrt{B^2 - B_z^2}$$

Only the component transverse to the Sun–Earth line is retained for the Newell coupling function.

Newell coupling proxy.

$$\Phi_N = v^{(4/3)} B_t^{(2/3)} \sin^{(8/3)}(\theta/2)$$

This is a widely used empirical solar-wind–magnetosphere coupling measure.

Kan–Lee electric-field proxy.

$$E_{KL} = 0.33 (v/1000) B \sin^3(\theta/2)$$

The implementation uses the code’s own normalization and is best interpreted as a scaled coupling proxy.

Injection function.

$$Q = 4.4 (0.001 v |B_z| - 0.5), \text{ for } B_z < 0$$

Only southward IMF contributes to the storm-injection proxy.

Decay time.

$$\tau = 2.4 \exp[9.74 / (4.69 + Q)]$$

The decay timescale is computed from the same storm-injection proxy.

Estimated Dst response.

$$Dst_{est} = -Q \tau [1 - \exp(-dur / \tau)]$$

This finite-duration solution approximates the main-phase storm response.

Plasmapause L-shell.

$$L_{pp} = \max(1.8, 5.6 - 0.46 Kp)$$

The plasmapause is moved inward under elevated geomagnetic activity.

Equatorial field at plasmapause.

$$B_{pp} = B_E / L_{pp}^3$$

A dipole scaling estimates the field magnitude at the plasmapause.

Proton cyclotron frequency.

$$f_{pc1} = q_e B_{pp} / (2\pi m_p)$$

This is the natural Pc1 estimate before audio rescaling.

Pc5 / field-line resonance proxy.

$$f_{pc5} \approx v_{A,inner} / (4 L_{pp} R_E)$$

The code assumes inner-magnetospheric Alfvén speed is ten times the solar-wind Alfvén speed.

Schumann resonance proxy.

$$f_n = [c / (2\pi R_E)] \sqrt{n(n+1)}$$

The code applies a small positive correction as xray increases.

Kelvin–Helmholtz proxy.

$$f_{KH} \approx (v_{sw} - v_{ms}) / (2 R_{mp} R_E)$$

A highly simplified surface-wave growth-rate proxy.

Magnetopause erosion metric.

$$R_{erosion} = 11.4 - R_{mp}$$

A diagnostic measure of inward displacement relative to nominal conditions.

Scientifically, this function package should be interpreted as a reduced-order state-diagnostics module rather than a full magnetohydrodynamic solver.

Worked Example: February 22, 2024 Storm

Using the event record encoded for the February 22, 2024 storm ($v = 650$ km/s, $n = 18$ cm⁻³, $B = 55$ nT, $B_z = -32$ nT, $K_p = 8.3$, $Dst = -219$ nT, $xray = 6.3 \times 10^{-4}$ W/m², $dur = 18$ h), the implemented formulas produce the following representative diagnostics.

Table 2. Representative outputs from the implemented formulas for the February 22, 2024 event.

Quantity	Value	Interpretation
P_dyn	6.36 nPa	Strong dynamic pressure and significant dayside compression
R_mp	8.61 R_E	Compressed magnetopause

		relative to quiet conditions
R_bs	11.71 R_E	Bow shock displaced earthward with the magnetopause
v_A	282.76 km/s	Elevated Alfvén speed under strong field conditions
M_A	2.30	Moderately super-Alfvénic flow
Φ_N	88.05 (code units)	Substantial coupling potential
E_KL	1.13 (code units)	Elevated reconnection proxy
Dst_est	-237.49 nT	Storm-strength estimate comparable to encoded Dst
L_pp	1.80	Plasmapause compressed to the code's lower bound
f_pc1	80.25 Hz	Cyclotron-derived value prior to any additional scaling
f_pc5	61.64 mHz	Very low frequency used for modulation
f_sch1	7.93 Hz	Slightly elevated Schumann fundamental
f_KH	0.00242 mHz	Very low Kelvin–Helmholtz-rate estimate

Temporal Interpolation Function

The function `computeTimeP(p)` is a physical approximation model instead of a full storm-evolution solver. In the implemented rule set, B is scaled by $(1 - 0.3f)$, B_z by $(1 - f)$, K_p by $(1 - 0.5f)$, and Dst by $(1 - f)$, while v and n are held constant. The scientific meaning of this operation is limited: it approximates recovery by progressively reducing geoeffective magnetic forcing and geomagnetic response without explicitly evolving the solar-wind plasma state.

Sonification Functions

The sonification function `buildGraph(ctx, p, ph)` constructs a seven-layer audio graph. Each layer is tied to a physical quantity or proxy quantity. Because the mapping converts sub-

audio or supra-physical scales into audible bands, the sonification is scientifically structured but not physically literal.

Table 3. Audio-layer mappings used by the sonification function.

Layer	Implemented mapping	Scientific meaning
1. Ring current	$f_{\text{ring}} = 22 + 58(\text{Dst} /1600)^{0.5}$; amplitude $\propto \text{Dst} $	Storm-time ring-current intensity is represented as a low-frequency drone.
2. Alfvénic fundamental	$f_{\text{alf}} = 80(600/80)^{((v_A - 10)/590)}$; distortion $\propto \Phi_N$	Increasing Alfvén speed and coupling strengthen the central tonal component and nonlinear distortion.
3. Harmonic stack	Number of overtones $\approx \text{round}(K_p) + 1$	Greater geomagnetic activity increases spectral complexity.
4. Pc1 layer	$f_{\text{pc1, audio}} = 80 f_{\text{pc1}}$, clipped to 30-600 Hz; LFO rate $\propto K_p$	Cyclotron-frequency information is translated into an audible band and then amplitude-modulated.
5. Schumann tones	$f_{\text{sch1}} \times 4, f_{\text{sch2}} \times 2, f_{\text{sch3}} \times 1.5$	Global cavity resonances become low and mid-frequency reference tones.
6. FM synthesis	Carrier = $1.5 f_{\text{alf}}$; modulation rate $\propto f_{\text{pc5}}$; FM depth $\propto B_z /B$	Southward IMF and field-line dynamics modulate spectral sideband structure.
7. Hiss / KH modulation	Noise amplitude $\propto n$; band-pass center = $300 + 90K_p$; master LFO $\propto f_{\text{KH}}$	Turbulent texture encodes density, activity, and boundary-instability rate.

Several details are especially important scientifically. First, the source code rescales natural frequencies into the audio band. Second, storm severity is encoded not only through pitch but also through amplitude, distortion, modulation depth, and spectral complexity. Third, the offline export routine `renderWAV()` renders a fixed 30 s stereo file at 44.1 kHz, which standardizes output length across events and permits direct comparison between storms.

Visualization Functions

The canvas function `draw()` uses the same physics object `ph` to control scene geometry. The magnetopause and bow-shock radii are converted to screen coordinates through $R_{mp,px} = (R_{mp} / 12)H^{0.42}$ and $R_{bs,px} = (R_{bs} / 12)H^{0.42}$, where H is the canvas height. Particle count is initialized as approximately $8 + 1.2n$, so increasing density yields a more crowded upstream flow field. The bow shock is drawn as a dashed conic section, while the magnetotail is rendered as a tapered sheet and central current layer.

The function also derives a pulse term from the current audio level when playback is active. Consequently, the visualization is not only physics-driven but also acoustically coupled. The scientific interpretation of that coupling is representational rather than causal.

Export and Utility Functions

Not all functions in the source are physical-model functions. `encodeWAV(left, right, sr)` serializes floating-point stereo audio into a 16-bit PCM RIFF/WAVE buffer. `stopAudio()`, `startAudio()`, `startRecord()`, `stopRecord()`, and `exportWAV()` manage browser audio state, recording control, and offline file generation.

Table 4. Principal functions and their scientific or infrastructural roles.

Function	Category	Role in experiment
<code>computePhysics</code>	physics	Transforms event variables into derived space-physics diagnostics.
<code>computeTimeP</code>	timeline	Applies a nonphysical interpolation to selected storm variables.
<code>buildGraph</code>	sonification	Constructs the Web Audio graph and seven-layer mapping.
<code>draw</code>	visualization	Renders the magnetospheric scene from the derived state.
<code>renderWAV</code>	export	Performs a fixed-duration

		offline audio render.
encodeWAV	serialization	Writes stereo PCM audio into a RIFF/WAVE container.
startAudio / stopAudio	control	Starts or destroys the live audio graph.
startRecord / stopRecord / exportWAV	recording	Handles browser-mediated recording and download.

Experimental Interpretation and Limitations

The experiment can be understood scientifically as a physically informed multimodal encoding of CME consequences at Earth. Its strongest methodological feature is transparency: the primary variables, intermediate diagnostics, and output mappings are explicit in the source. However, the experiment uses event-level scalar summaries rather than time-resolved solar-wind streams, magnetic-field vectors, or particle fluxes. Some formulas are simplified or normalized for interactive use, the timeline function is graphical rather than dynamical, and historical benchmark events carry explicit uncertainty labels.

The correct scientific description is therefore the following: SON is an empirical, reduced-order, parameter-mapped sonification and visualization experiment for CME-driven space-weather events. It uses published magnetospheric scaling relations and code-specific transfer functions to convert a compact event-state vector into audio-visual outputs that preserve relative physical severity, coupling strength, and boundary compression.

Conclusion

The formulas and functions in SON form a coherent computational experiment. The event database supplies a compact representation of solar-wind and geomagnetic forcing. The computePhysics() function converts those inputs into compression, coupling, resonance, and storm-response diagnostics. The buildGraph() function maps those diagnostics into layered

audio, while `draw()` maps them into boundary geometry and particle motion. Utility functions then export the result as standardized stereo audio. The system is scientifically meaningful because its mappings are explicit and physically ordered, but it remains a reduced-order interpretive model rather than a first-principles simulation.

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